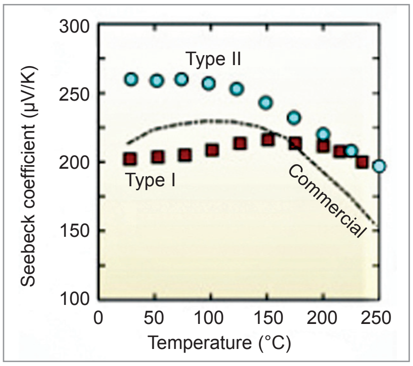

Fig. 7: Seebeck coefficient of improved Type I and Type II BiTe material

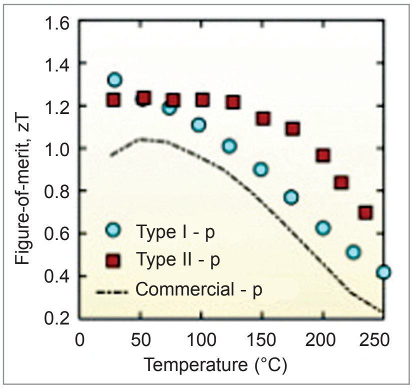

Fig. 8: Figure of merit of improved BiTe material

With improved techniques, p-type BiTe TE material that has average ZT of 1.08 and n-type BiTe with 0.84 ZT has been made. The significant enhancement in ZT could be achieved through compositional and defect engineering.

Type I module is constructed using p-type $Bi_{0.5}Sb_{1.5}Te_3$ and n-type using $Bi_2Te_{2.7}Se_{0.3}S_{0.01}$. Type II material is constructed from p-type $Bi_{0.4}Sb_{1.6}Te_3$ and n-type $Bi_2Te_{2.7}Se_{0.3}S_{0.01}Cu_{0.01}$ materials. Fig. 7 and Fig. 8 show the Seebeck coefficient and figure of merit of these two types of improved BiTe materials, respectively.

2. Lead telluride (PbTe) alloy. Low conversion efficiency is a big obstacle that impedes large scale application of TE materials for power generation. Lead telluride alloy is recognised as an excellent compound for power generation in the mid temperature range of

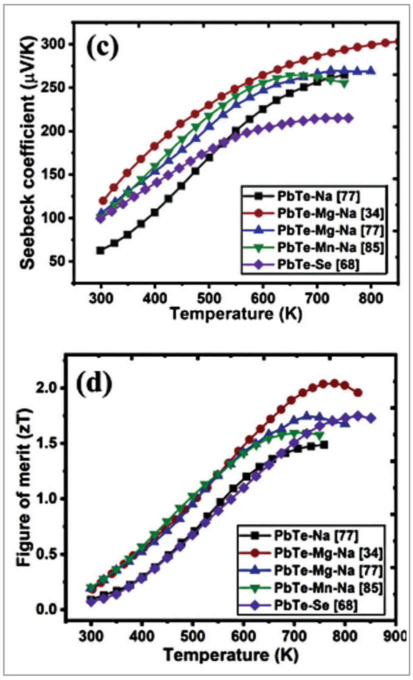

Fig. 9: Seebeck coefficient and ZT of lead telluride TE material

500-800°K. It has highly symmetric rock salt crystal structure, which is chemically and thermally stable. Lead telluride can be made either a p-type or n-type semiconductor.

The last decade has witnessed a quantum jump in TE properties of PbTe. Its maximum figure of merit (ZT) has already crossed 2. It has a unique combination of high power factor and significantly lower thermal conductivity, high Seebeck coefficient, and high electrical conductivity. Fig. 9 shows Seebeck coefficient and figure of merit of PbTe materials recently developed. It has a drawback that it is highly toxic in nature. Commercial PbTe material has about 7% conversion efficiency with ZT about 1.0.

Recently researchers have enhanced the ZT of a sintered material to 1.8 (600°C) using a nanostructure forming technology. Further, an electrode material has been developed that contacts very well electrically and thermally with PbTe containing MgTe nanostructures, achieving a conversion efficiency of about 11% with hot side at 600°C and the cold side at 10°C.

This breakthrough has made the way to convert waste heat and solar thermal power to large scale practical application. Because the nanostructures formed in the PbTe sintered material effectively scatter heat carrying phonons, but have no effect on the charge carrier transport, there is dramatic improvement in the ZT.

3. Silicon-germanium alloy. It is a kind of semiconductor that is often used for thermoelectricity generation with a working temperature around 1300°C. Combining Si and Ge allows to retain high electrical conductivity of both components and reduce the thermal conductivity due to the increased phonon scattering as Si and Ge have different lattice properties. Due to this reason Si-Ge alloys are currently the best TE material for high temperature application.

Conventional Si-Ge materials have ZT values of 0.9 and 0.5 at 1200K for n-type and p-type materials, respectively. By adopting nanocomposite approach reducing the grain size to around 5nm, ZT values can be increased by a factor of two and conversion efficiency can also be increased considerably. Ge is a scarce material and hence this alloy is costly. However, using only 5% of Ge it is possible to have Si-Ge alloy with improved performance.

The advantage of the newly developed nanocomposites is that its ZT values consistently remain above 1 over high temperature range between 600°C and 1000°C. Hence, these are the best TE materials for high temperature application for power generation by solar radiation, radioisotope devices, and waste heat recovery system. Table 2 shows the ZT values of bulk and improved nanostructured Si-Ge TE materials.

A thermoelectric device with a ZT of 1.25 will have an efficiency of about 10%. A segmented Si-Ge TE device over temperature range of